

Closing the Loop Inside Neural Networks: Causality-Guided Layer Adaptation for Fault Recovery Control

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Abstract—This letter studies the problem of real-time fault recovery control for nonlinear control-affine systems subject to actuator loss of effectiveness faults and external disturbances. We develop a two-stage framework that combines causal inference with selective online adaptation to achieve an effective learning-based recovery control method. In the offline phase, we introduce a causal layer attribution technique based on the average causal effect (ACE) to evaluate the relative importance of each layer in a pretrained deep neural network (DNN) controller compensating for faults. This provides a principled approach to select the most causally influential layer for fault recovery control in the sense of ACE, and goes beyond the widely used last-layer adaptation approach. In the online phase, we deploy a Lyapunov-based gradient update to adapt only the ACE-selected layer to circumvent the need for full-network or last-layer only updates. The proposed adaptive controller guarantees uniform ultimate boundedness (UUB) with exponential convergence of the closed-loop system in the presence of actuator faults and external disturbances. Compared to conventional adaptive DNN controllers with full-network adaptation, our methodology has a reduced computational overhead in the online phase. To demonstrate the effectiveness of our proposed methodology, a case study is provided on a 3-axis attitude control system of a spacecraft with four reaction wheels.

Index Terms—Fault recovery control, adaptive control, online learning, adaptive neural networks.

I. INTRODUCTION

AUTONOMOUS systems ranging from unmanned aerial vehicles (UAV) to spacecraft are required to have high levels of resilience to operate safely in dynamic and uncertain environments [1], [2]. These systems frequently encounter challenges such as unpredictable external disturbances and various types of faults, which push them into out-of-distribution (OOD) conditions beyond their nominal design assumptions [3]. Under such circumstances, conventional controllers

often fail to recover the system since they lack the level of adaptability required to maintain a reliable performance.

Adaptive control methods [4], [5] offer strong stability guarantees when dealing with structured uncertainty or known fault models. However, their effectiveness depends critically on accurate system modeling and prior knowledge of fault characteristics. Data-driven controllers that leverage deep neural networks (DNNs) provide a promising alternative by capturing complex mappings directly from data. However, their performance declines when faced with OOD scenarios during runtime [6]. Hence, online learning approaches where neural network parameters are updated online have been employed as a solution to make a closed-loop autonomous system robust to modeling uncertainties and faults [7], [8].

Spectral normalization, which constrains the Lipschitz constant of each DNN layer by limiting its spectral norm, has been shown to improve generalization under distribution changes by controlling layer-wise sensitivity [9]. This structural regularization has inspired adaptive strategies that emphasize layer-wise DNN adjustments during online operation. For instance, [10] utilizes spectral normalization and a meta-learning framework to identify and learn disturbance features offline and update only the final layer of DNN online for rapid residual learning and disturbance rejection.

In the control systems domain, hybrid adaptive methods are utilized to estimate uncertain dynamics via Bayesian learning combined with control-barrier functions [11]. In [12], a long-short-term memory (LSTM)-based identifier whose weights are adapted online under Lyapunov guarantees has been employed. Sparse adaptation techniques utilizing L_1 regularization have been proposed to detect actuator faults and reconfigure controllers [1]. More classical fault-tolerant adaptive control laws update parameters continuously to accommodate various actuator faults [13]. Although these approaches span a wide range of strategies, they treat the DNN-based controller as a single block. In contrast, our framework incorporates causal inference and average causal effect (ACE) to determine which subset of DNN layers is the most critical for fault recovery.

In this letter, first, a nominal controller is designed for the system with no actuator fault and external disturbance. Consequently, we train a DNN controller offline to compensate for a wide range of fault and disturbance scenarios that the nominal controller cannot cope with. However, since our DNN

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controller may still encounter actuator faults and disturbances for which it was not trained, we introduce a method to identify and adapt only the most causally influential layer of the network. In the offline phase, we simulate various actuator failure and degradation conditions under external disturbances to train the DNN controller in a supervised manner and similar to the imitation learning (IL) approach [2]. Consequently, we compute and use the ACE to measure how changes in each layer's weights affect the closed-loop tracking error. This step allows us to identify which layer has the most significant causal impact on the system's performance under fault conditions. Although the ACE evaluation can be computationally expensive, it is executed offline and before deployment.

In the online phase, only the layer identified by the ACE is updated using an adaptive control law that ensures the tracking error remains uniformly ultimately bounded (UUB) despite unknown faults and disturbances (see Fig. 1). Subsequently, we investigate and study the case where a fault detection and identification (FDI) module can provide an estimate of the fault. The main contributions of this letter are: (1) We introduce a structural causal model (SCM) for the DNN-based controller and develop a new methodology to compute the ACE of each layer on the closed-loop tracking error; (2) we develop an adaptation law with exponential stability that updates only the ACE-selected layer, which reduces the computational overhead in the online phase and avoids full-network updates. We prove that this selective adaptation guarantees the UUB of the tracking error; (3) we integrate causal inference with adaptive control to identify and adapt an internal DNN layer (beyond the last-layer adaptation in [1], [10]).

II. PROBLEM FORMULATION

A. Notations

For a matrix A , $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote its minimum and maximum eigenvalues, respectively. The operator $\text{diag}(\cdot)$ forms a diagonal matrix; $\text{tr}(\cdot)$ denotes the trace; $\text{Im}(\cdot)$ is the image (column space); and $A^\dagger$ represents a bounded right-inverse (i.e., the Moore-Penrose pseudo-inverse restricted to $\text{Im}(A)$). The symbol $\|\cdot\|$ denotes the Euclidean 2-norm; $\|\cdot\|_F$ denotes the Frobenius norm. Finally, $\|\cdot\|_{\text{Lip}}$ represents the global Lipschitz constant of a map.

B. System Model

We consider a nonlinear control-affine system as

$$\dot{x} = f(x) + g(x) \Lambda u + d, \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input, and $d \in \mathbb{R}^n$ is a bounded disturbance, i.e., $\|d\| \leq \bar{d}$, where $\bar{d} > 0$ is an unknown upper bound. Also, $f(x)$ and $g(x)$ are known smooth functions. Moreover, $\Lambda = \text{diag}(\eta_1(t), \dots, \eta_m(t))$ is an unknown diagonal matrix, where $0 < \eta_k(t) \leq 1$ denotes a loss of effectiveness fault in the k -th actuator, for $k = 1, \dots, m$. In this letter, we consider $\eta_{\min} \leq \eta_k \leq 1$, where $\eta_{\min} > 0$ is an unknown value that indicates the minimum effectiveness of the actuator, and $\eta_k = 1$ indicates that the k -th actuator

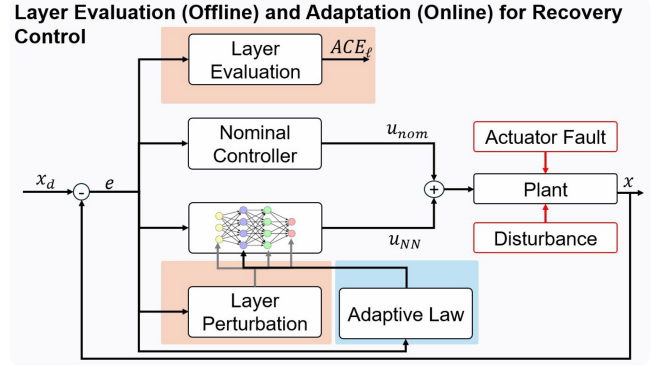


Fig. 1. Layer importance evaluation (shaded orange) and updating the selected layer (shaded blue).

is fault-free or healthy. Also, we assume that $d \in \text{Im}(g(x))$, which implies that (1) has a matched disturbance.

C. Controller Design and Tracking Error

Consider (1) under fault-free and disturbance-free conditions, i.e., $\Lambda = I$ and $d = 0$. A nominal control law can be designed as $u_{\text{nom}} = g(x)^\dagger (\dot{x}_d - f(x) - Ke)$, where $e = x - x_d$ is the tracking error and $x_d \in \mathbb{R}^n$ is the desired trajectory. We choose (x_d, K) so that $\dot{x}_d - f(x) - Ke \in \text{Im}(g(x))$ for all t . Moreover, the control gain $K > 0$ guarantees $\dot{V}_0(e) \leq -\alpha_1(\|e\|)$ under nominal conditions, i.e., $\Lambda = I$ and $d = 0$, where $V_0(e) = \frac{1}{2}e^\top Pe$ is a Lyapunov candidate function, $P > 0$ and symmetric, and $\alpha_1(\cdot)$ is a class- $\mathcal{K}$ function [14].

Under nominal conditions, having $u = u_{\text{nom}}$ guarantees the exponential stability of the closed-loop system [14]. However, this controller cannot guarantee the boundedness of the tracking error e in the presence of faults and disturbances. Hence, we design a DNN-based controller $u_{\text{NN}}(e, u_{\text{nom}}; \bar{W})$ that can compensate for the impact of the actuator fault and the disturbance in the closed-loop system such that the total control input is $u = u_{\text{nom}} + u_{\text{NN}}$, where $\bar{W}$ denotes the design parameters of the DNN.

D. Problem Statement

Given u_{NN} , we identify the update of which DNN layer can result in minimization of tracking error in the presence of unknown faults and disturbances. The second problem is to find an adaptive law that can be used to update the weights of the identified layer in the previous step. Moreover, the adaptive law should guarantee the uniform ultimate boundedness of the tracking error.

III. OFFLINE PHASE: DNN TRAINING AND LAYER IMPORTANCE EVALUATION

A. Data Generation and Supervised DNN Training

A rich dataset of fault and disturbance scenarios is needed to train the u_{NN} in a supervised manner to develop a baseline fault compensator. Hence, we sample M scenarios $\{\hat{\Lambda}^{(i)}, d^{(i)}\}_{i=1}^M$, where $\hat{\Lambda}^{(i)} = \text{diag}(\eta_1^{(i)}, \dots, \eta_m^{(i)})$ with each $\eta_{\min} < \eta_k^{(i)} \leq 1$ covering light to severe actuator degradation, and $d^{(i)}$ is a bounded disturbance. The i -th fault initiation time $t_f^{(i)}$ is

selected randomly. Moreover, the desired trajectory $x_d^{(i)}$ can be designed based on our control tracking objectives, and the initial conditions are randomized.

For each scenario i , we have $\dot{x}^{(i)} = f(x^{(i)}) + g(x^{(i)})\hat{\Lambda}^{(i)}[u_{\text{nom}}^{(i)} + u_{\text{comp}}^{(i)}] + d^{(i)}$, where $x^{(i)} \in \mathbb{R}^n$, $u_{\text{nom}}^{(i)}$ is the nominal controller for the i -th scenario as in Section II-C, and $u_{\text{comp}}^{(i)}$ is an ideal controller that compensates for the impact of faults and disturbances. The u_{comp} is employed only during the offline phase to generate expert demonstrations that capture actuator fault and disturbance effects beyond the nominal controller's capability.

One has $\dot{x}^{(i)} = f(x^{(i)}) + g(x^{(i)})u_{\text{nom}}^{(i)} + g(x^{(i)})u_{\text{comp}}^{(i)} + g(x^{(i)})[\hat{\Lambda}^{(i)} - I]u^{(i)} + d^{(i)}$, where $u^{(i)} = u_{\text{nom}}^{(i)} + u_{\text{comp}}^{(i)}$. Since $d \in \text{Im}(g(x^{(i)}))$, one can always find a certain $\hat{d}^{(i)}$ such that $d^{(i)} = g(x^{(i)})\hat{d}^{(i)}$. Hence, the ideal compensator can be designed as $u_{\text{comp}}^{(i)} = -(\beta^{(i)}u^{(i)} + \hat{d}^{(i)})$, where $\beta^{(i)} = \hat{\Lambda}^{(i)} - I$. Considering that each element of the diagonal matrix $\beta^{(i)}$ is greater than -1 and less than or equal to 0 , the matrix $I + \beta^{(i)}$ is invertible. Thus, one obtains $u_{\text{comp}}^{(i)} = -[I + \beta^{(i)}]^{-1}(u_{\text{nom}}^{(i)} + \hat{d}^{(i)})$. Considering $e^{(i)} = x^{(i)} - x_d^{(i)}$, $u_{\text{nom}}^{(i)}$, and $u_{\text{comp}}^{(i)}$, the training dataset $\mathcal{D}_{\text{train}} = \{(e^{(i)}, u_{\text{nom}}^{(i)}, u_{\text{comp}}^{(i)})\}_{i=1}^M$ can be created.

Consequently, one can train $u_{\text{NN}}(e, u_{\text{nom}}; \bar{W})$ by minimizing the loss function $\mathcal{L}_{\text{train}}(\bar{W}) = \frac{1}{M} \sum_{i=1}^M \frac{1}{T_i} \int_0^{T_i} (\|u_{\text{NN}} - u_{\text{comp}}^{(i)}\|^2 + \lambda_e \|\hat{e}^{(i)}\|^2)$, where $[0, T_i]$ denotes the time window of the i -th scenario, and $\hat{e}^{(i)} = f(x^{(i)}) + g(x^{(i)})\hat{\Lambda}^{(i)}[u_{\text{nom}}^{(i)} + u_{\text{NN}}] + d^{(i)} - \dot{x}_d^{(i)}$. We also utilize the spectral normalization method [15] to enforce a global Lipschitz bound on the u_{NN} . If the activation function of the DNN has a Lipschitz constant equal to 1 , one obtains $\|u_{\text{NN}}\|_{\text{Lip}} \leq \|W_1, \dots, W_L\| \leq \prod_{\ell=1}^L \|W_\ell\|$, where W_ℓ is the ℓ -th weight matrix of the DNN with appropriate dimensions, for each $\ell = 1, \dots, L$. Hence, enforcing $\|W_\ell\| < 1$ for every layer results in having $\|u_{\text{NN}}\|_{\text{Lip}} \leq 1$.

The DNN training procedure described above is similar to the behavioral cloning in imitation learning (IL) [2], [16]. The analytically derived compensator u_{comp} serves as an expert policy, the collected tuples $(e^{(i)}, u_{\text{nom}}^{(i)}, u_{\text{comp}}^{(i)})$ constitute the set of demonstrations, and the quadratic loss $\mathcal{L}_{\text{train}}(\bar{W})$ minimizes the error between the learner u_{NN} and the expert. A distribution shift or mismatch (e.g., unseen fault and disturbance scenarios) can arise when the learner, i.e., u_{NN} , is deployed. Hence, in the next subsection, the ACE-based method is proposed to find the parameters that the learner can update in response to the mentioned distribution shifts. Let $\zeta = [u_{\text{nom}}^\top e^\top]^\top$. The DNN structure can be shown as $z_0 = \zeta$, $z_\ell = \phi_\ell(a_\ell)$, $a_\ell = W_\ell z_{\ell-1} + b_\ell$, for all $\ell = 1, \dots, L-1$, where ϕ_ℓ is a smooth activation function (e.g., tanh, sigmoid, softplus), $a_\ell \in \mathbb{R}^{n_\ell}$, $b_\ell \in \mathbb{R}^{n_\ell}$ is the bias, $z_{\ell-1} \in \mathbb{R}^{n_{\ell-1}}$, and $W_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$. The output of the DNN-based controller is $u_{\text{NN}} = -W_L z_{L-1}$.

B. Average Causal Effect (ACE) for Layer Importance Evaluation

In [17], a causal-attribution framework, which recasts a DNN as a structural causal model (SCM) is proposed to quantify the ACE of any neuron on the output. We adopt this approach and extend it from neuron-level explanations to layer-level importance evaluation for our closed-loop system. In particular, we utilize the do-operator (see [18]) to impose

interventions on each weight matrix W_ℓ . These interventions are used to perturb layers of the u_{NN} and measure how they alter our closed-loop tracking error $e = x - x_d$. By computing the expected change in $\|e\|$ caused by a small random offset in each layer, one can obtain an ACE score that directly measures how sensitive our error is to a change in layer ℓ . This causality-based approach offers a principled alternative to gradient-based attribution methods [19] to decide which layer to update and compensate for the impact of faults and disturbances. It should be noted that this step is carried out offline and before deploying u_{NN} .

To formalize the above, we represent the DNN-based controller as an SCM. Let the set of nodes $\mathcal{V} = \{z_0(t), \dots, z_L(t), u_{\text{NN}}(t), e(t + \Delta t)\}$ with directed edges in $\mathcal{E}$ that follow the cascade $z_0(t) \rightarrow z_1(t) \rightarrow \dots \rightarrow z_L(t) \rightarrow u_{\text{NN}}(t) \rightarrow e(t + \Delta t)$, where $\Delta t > 0$ denotes a small time increase in the continuous-time evolution, such that causal ordering is interpreted in a discrete-time unrolled sense and variables at time t influence the state and tracking error at the next instant $t + \Delta t$. In this representation, the weights W_ℓ parameterize the edges and interventions are written as $\text{do}(W_\ell \mapsto W_\ell + \Delta_\ell)$. This demonstrates how perturbations propagate from layer ℓ to the error e .

Let us define the fault, disturbance, and desired trajectory triplet $(\check{\Lambda}, \check{d}, \check{x}_d)$ that was not used to train the u_{NN} in Section III-A. For each layer ℓ , we define the ACE as

$$\text{ACE}_\ell := \mathbb{E}[\|\check{e}\| \mid \text{do}(W_\ell + \Delta_\ell(t))] - \mathbb{E}[\|\check{e}\| \mid W_\ell], \quad (2)$$

where $\mathbb{E}[\|\check{e}\| \mid \text{do}(W_\ell + \Delta_\ell(t))]$ is the expected value of the error $\check{e} = \check{x} - \check{x}_d$, and $\check{x}$ is the state generated using (1) but the actuator fault is set to $\check{\Lambda}$, the disturbance is $\check{d}$, and $\Delta_\ell(t)$ is added to the weight matrix of the ℓ -th layer of the u_{NN} . Moreover, $\text{do}(W_\ell + \Delta_\ell(t))$ denotes a causal intervention that perturbs layer ℓ by a small weight offset $\Delta_\ell(t)$, i.e., the weight of the ℓ -th layer becomes $W_\ell + \Delta_\ell(t)$ (see [18] for more details on the do-operator). We set $\Delta_\ell(t) \sim \mathcal{N}(0, \rho_\ell^2 I)$, where ρ_ℓ is the standard deviation of the perturbation for the ℓ -th layer. A negative ACE_ℓ means that intervening on ℓ decreases the tracking error. Thus, the ℓ -th layer with the smallest ACE_ℓ is causally important for compensating for out-of-distribution (OOD) fault and disturbance scenarios. One can also evaluate (2) for various OOD scenarios of $(\check{\Lambda}, \check{d}, \check{x}_d)$ and average the resulting ACE_ℓ to generalize and improve the ACE-based layer importance evaluation.

Monte Carlo methods are widely used to approximate intractable quantities [20], such as ACE_ℓ . Since the exact computation of (2) is challenging, one can approximate it by using a Monte Carlo estimation as $\widehat{\text{ACE}}_\ell = \frac{1}{N} \sum_{j=1}^N \|e_j(W_\ell + \Delta_\ell^{(j)})\| - \|e_j(W_\ell)\|$, where we sample N random perturbations $\Delta_\ell^{(j)}$ and simulate the closed-loop error under the corresponding faults and disturbances. Consequently, we select a layer with the minimum $\widehat{\text{ACE}}_\ell$ as the best (in the sense of the ACE) to update in the online step.

Table I summarizes various strategies for selecting which DNN layer to adapt during the online step for fault recovery control. These approaches differ in how they quantify layer importance and range from fixed structural rules and local sensitivity metrics to magnitude-based heuristics. In contrast, our proposed ACE-based strategy provides a principled, causal

TABLE I
COMPARISON OF LAYER EVALUATION STRATEGIES

Method	Description	Main Limitations
Last-Layer-Only [10], [21]	Updates only the final layer of the DNN during online operation. Assumes early layers retain valid representations across all faults and disturbances.	Ignores potential inaccuracy of internal representations; may fail under severe or multiple actuator faults.
Jacobian-Based Sensitivity [19]	Selects the layer with highest local sensitivity, e.g., $\ \frac{\partial u_{NN}}{\partial W_\ell}\ $ or $\ \frac{\partial \mathcal{L}}{\partial W_\ell}\ $. Measures instantaneous influence of weights on the DNN's output or loss.	Captures only short-term correlations; neglects closed-loop dynamics and long-horizon fault effects.
Weight-Magnitude [22]	Ranks neurons in each layer based on $\ W_\ell\ $ or $ W_\ell $ to approximate contribution of each layer.	Static and does not reflect how layer perturbations propagate through the system dynamics.
Our ACE-Based	Computes ACE_ℓ in (2) to quantify the causal impact of each layer on the closed-loop error.	Requires offline Monte Carlo evaluation.

criterion that identifies the layer with the greatest impact (in the sense of (2)) on the closed-loop tracking error to be updated for fault recovery.

Remark 1: The definition in (2) uses $\|\tilde{e}\|$ as the performance metric. Alternatively, one can adopt a Lyapunov-based measure that ties ACE directly to the closed-loop stability. In particular, we can define $ACE_\ell^V = \mathbb{E}[\dot{V}_0 | \text{do}(W_\ell + \Delta_\ell(t))] - \mathbb{E}[\dot{V}_0 | W_\ell]$. This definition of the ACE quantifies how interventions in layer ℓ affect the Lyapunov drift and provides a stability-oriented criterion for layer selection.

IV. ONLINE ADAPTATION OF THE DNN FOR FAULT RECOVERY CONTROL

From Section III, one can identify which layer in the DNN should be updated online to compensate for actuator faults and disturbances, as shown in Fig. 1. In this section, we first derive an appropriate adaptive law for u_{NN} , and then show that the aforementioned adaptive law guarantees UUB of the tracking error. Finally, we show that if there is a fault detection and identification (FDI) module that can provide an estimate of fault values, that is, Λ , the error bounds would decrease.

A. Adaptive Law for a Selected Layer

Suppose $\ell^* \in \{1, \dots, L\}$ is the selected layer in the previous section to compensate for OOD scenarios. Substituting $u = u_{nom} + u_{NN}$ in (1), for the error dynamics one obtains

$$\dot{e} = [f(x) + g(x)u_{nom} - \dot{x}_d] + g(x)(u_{NN} + r), \quad (3)$$

where $r = [\Lambda - I]u + \hat{d}$ and $d = g(x)\hat{d}$.

Assumption 1: Let $\ell^* \neq L$. There exists a bounded and constant ideal weight $W_{\ell^*}^* = W^*$ such that $r = W_L z_{L-1}^* + \epsilon(t)$, where z_{L-1}^* captures the impact of the ideal weight in the ℓ^* -th layer, i.e., $W_{\ell^*}^*$, and $\epsilon(t)$ is the bounded approximation error satisfying $\|\epsilon(t)\| \leq \bar{\epsilon}$. If $\ell^* = L$, we assume $r = W^* z_{L-1} + \epsilon(t)$.

It should be noted in Assumption 1, W^* is the ideal reference weight toward which the adaptive law drives the estimated weights W_{ℓ^*} , thereby ensuring minimization of the tracking error within a certain bound (see Theorem 1). The existence of such an ideal weight W^* is a standard assumption

in adaptive control and neural network-based approximation frameworks, and is commonly employed to establish convergence and boundedness properties [10], [14], [21].

One needs to come up with an adaptive law for the DNN controller such that $u_{NN} \approx -W_L z_{L-1}^*$ if $\ell^* \neq L$ and $u_{NN} \approx -W_L^* z_{L-1}$, otherwise. Therefore, we derive the adaptive law for both cases of $\ell^* \neq L$ and $\ell^* = L$, below. Let us consider the virtual loss function $\mathcal{L} = \delta_L^\top u_{NN}$ for updating the parameters of u_{NN} , where $\delta_L = g(x)^\top P e \in \mathbb{R}^m$. Moreover, for $\ell = L-1, \dots, 1$, we define $\delta_\ell = \Psi_\ell(a_\ell) W_{\ell+1}^\top \delta_{\ell+1}$, where $\Psi_\ell = \text{diag}(\psi'_{1,\ell}(a_{1,\ell}), \dots, \psi'_{n_\ell,\ell}(a_{n_\ell,\ell}))$ is the Jacobian of the activation function. Hence, one obtains $\delta_{\ell^*} = \Psi_{\ell^*} W_{\ell^*+1}^\top \dots \Psi_{L-1} W_L^\top \delta_L$, where δ_{ℓ^*} is the backpropagated error to the ℓ^* -th layer.

The gradient of the virtual loss function with respect to the adjustable weight matrix is $\nabla_{W_{\ell^*}} \mathcal{L} = -\delta_{\ell^*} z_{\ell^*-1}^\top$. Therefore, the adaptive law for updating the layer $\ell^* \in \{1, \dots, L\}$ can be derived in the following form:

$$\dot{W}_{\ell^*} = \Gamma \delta_{\ell^*} z_{\ell^*-1}^\top - \gamma W_{\ell^*}, \quad (4)$$

where $\Gamma > 0$ is a diagonal gain matrix and $\gamma > 0$.

B. Exponential Stability and Robustness Analysis

Let us define $\tilde{W} = W_{\ell^*} - W_{\ell^*}^*$. Consider the case where $\ell^* \neq L$. The first-order Taylor expansion of $z_{L-1} - z_{L-1}^*$ with respect to W_{ℓ^*} can be written as $z_{L-1} - z_{L-1}^* = [\Psi_{L-1} W_{L-1} \Psi_{L-2}, \dots, W_{\ell^*+1} \Psi_{\ell^*}] \tilde{W} z_{\ell^*-1}^* + \mathcal{O}(\|\tilde{W}\|^2)$, where $\mathcal{O}(\|\tilde{W}\|^2)$ denotes the higher order terms. Consequently, premultiplying $z_{L-1} - z_{L-1}^*$ by $\delta_L^\top W_L$ yields $\delta_L^\top W_L S \tilde{W} z_{\ell^*-1}^* + \mathcal{O} = \text{tr}(\tilde{W}^\top \delta_{\ell^*} z_{\ell^*-1}^\top) + \mathcal{O}$, where $S = \Psi_{L-1} W_{L-1} \Psi_{L-2}, \dots, W_{\ell^*+1} \Psi_{\ell^*} \in \mathbb{R}^{n_{L-1} \times n_{\ell^*}}$.

Consider the following Lyapunov candidate function $V(e, \tilde{W}) = V_0(e) + \frac{1}{2} \text{tr}(\tilde{W}^\top \Gamma^{-1} \tilde{W})$, where $V_0(e) = \frac{1}{2} e^\top P e$. The time derivative of V can be expressed by $\dot{V} = e^\top P \dot{e} + \text{tr}(\tilde{W}^\top \Gamma^{-1} \dot{\tilde{W}})$. Considering (3) and $Q = \frac{1}{2}(PK + K^\top P^\top)$, one has $\dot{V} = -e^\top Q e + \delta_L^\top (-W_L z_{L-1} + r) + \text{tr}(\tilde{W}^\top \Gamma^{-1} \dot{\tilde{W}})$. Hence, substituting r , $z_{L-1} - z_{L-1}^*$, and the adaptive law (4) yields

$$\dot{V} = -e^\top Q e - \gamma \text{tr}(\tilde{W}^\top \Gamma^{-1} W_{\ell^*}) + e^\top P g \epsilon + \mathcal{O}. \quad (5)$$

It should be noted that if $\ell^* = L$, $-W_L z_{L-1} + r$ above can be recast as $-W_L z_{L-1} + r = -\tilde{W} z_{L-1} + \epsilon$, and the remainder of the derivations for $\dot{V}$ follow along similar lines to the case of $\ell^* \neq L$. Moreover, the following stability proof is independent of the selected layer.

Assumption 2: For each hidden layer $\ell = 1, \dots, L$, the activation function ϕ_ℓ is continuously differentiable (i.e., $\mathcal{C}^1$) and bounded, such that $\|\phi_\ell(a_\ell)\| \leq \bar{\phi}$, where $\bar{\phi} > 0$ and the boundedness is guaranteed by spectral normalization. Moreover, the Jacobian of the activation function Ψ_ℓ satisfies $\|\Psi_\ell(a_\ell)\| \leq \bar{\Psi}$, where $\bar{\Psi} > 0$.

Theorem 1: Let Assumptions 1 and 2 hold. Considering the adaptive law (4), the tracking error e and weight error $\tilde{W}$ in the closed-loop system (1) with the total control input $u = u_{nom} + u_{NN}$ remain UUB with exponential convergence.

Proof: Let $\omega = \|\Gamma^{-\frac{1}{2}} \tilde{W}\|_F$ and $c = \|\Gamma^{-\frac{1}{2}} W^*\|_F$. One has $\gamma \|\Gamma^{-\frac{1}{2}} \tilde{W}\|_F \|\Gamma^{-\frac{1}{2}} W^*\|_F \leq \frac{\gamma}{2} \omega^2 + \frac{\gamma}{2} c^2$. Moreover, Young's

inequality results in having $\|P\| \|g\| \|e\| \bar{\epsilon} \leq \frac{\lambda_{\min}(Q)}{2} \|e\|^2 + \frac{\|P\|^2 \|g\|^2}{2\lambda_{\min}(Q)} \bar{\epsilon}^2$. Hence, from (5), we obtain $\dot{V} \leq -\frac{\lambda_{\min}(Q)}{2} \|e\|^2 - \frac{\gamma}{2} \omega^2 + \frac{\|P\|^2 \|g\|^2}{2\lambda_{\min}(Q)} \bar{\epsilon}^2 + \frac{\gamma}{2} c^2 + \bar{o}$, where $\mathcal{O}(\|\tilde{W}\|^2) \leq \bar{o}$.

From $V_0 \leq \frac{1}{2} \lambda_{\max}(P) \|e\|^2$, it can be inferred that $-\frac{\lambda_{\min}(Q)}{2} \|e\|^2 \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V_0$. Thus, $\dot{V} \leq -\alpha V + \sigma + \bar{o}$, holds for $\alpha = \min\{\frac{\lambda_{\min}(Q)}{2\lambda_{\max}(P)}, \frac{\gamma}{2}\}$ and $\sigma = \frac{\|P\|^2 \|g\|^2}{2\lambda_{\min}(Q)} \bar{\epsilon}^2 + \frac{\gamma}{2} \|\Gamma^{-\frac{1}{2}} W^*\|_F^2$. Since $\frac{\lambda_{\min}(P)}{2} \|e\|^2 \leq V$, one has $\|e(t)\| \leq e^{-\frac{\alpha t}{2}} \sqrt{\frac{2}{\lambda_{\min}(P)} V(0)} + \sqrt{\frac{2\sigma + \bar{o}}{\alpha \lambda_{\min}(P)}}$, and $\|\Gamma^{-1/2} \tilde{W}(t)\|_F \leq e^{-\frac{\alpha t}{2}} \sqrt{2V(0)} + \sqrt{\frac{2\sigma + \bar{o}}{\alpha}}$. ■

C. Improved Error Bounds With a Fault Detection and Identification (FDI) Module

Suppose a dedicated FDI module provides an online estimate $\hat{\Lambda}(t) = \text{diag}(\hat{\eta}_1(t), \dots, \hat{\eta}_m(t))$ of the actuator loss of effectiveness matrix $\Lambda(t) = \text{diag}(\eta_1(t), \dots, \eta_m(t))$, where $\hat{\eta}_{\min} < \hat{\eta}_k \leq 1$, and $\hat{\eta}_{\min} > 0$. We reconfigure the nominal controller to account for this estimate as $u_{\text{nom}}^{\text{FDI}} = [g(x) \hat{\Lambda}]^\dagger (\dot{x}_d - f(x) - Ke)$. Using $u = u_{\text{nom}}^{\text{FDI}} + u_{\text{NN}}$, the error dynamics is $\dot{e} = -Ke + g(x)(\Lambda u_{\text{NN}} + [\Lambda - \hat{\Lambda}] u_{\text{nom}}^{\text{FDI}} + \hat{d})$, where $d = g(x)\hat{d}$. Considering $g(x)\Lambda u_{\text{NN}} = g(x)(u_{\text{NN}} + [\Lambda - I]u_{\text{NN}})$, one obtains $\dot{e} = -Ke + g(x)(u_{\text{NN}} + r_{\text{FDI}})$, where $r_{\text{FDI}} = [\Lambda - I]u_{\text{NN}} + [\Lambda - \hat{\Lambda}] u_{\text{nom}}^{\text{FDI}} + \hat{d}$.

Assumption 3: There exist bounded constants $\bar{\Delta}_\Lambda \in (0, 1)$, $\kappa_g^\dagger > 0$, $\bar{v} > 0$, and $\bar{u}_{\text{NN}} > 0$ such that $\|\Lambda - \hat{\Lambda}\| \leq \bar{\Delta}_\Lambda$, $\|g(x)^\dagger\| \leq \kappa_g^\dagger$, $\|v(x)\| \leq \bar{v}$, and $\|u_{\text{NN}}\| \leq \bar{u}_{\text{NN}}$, $\forall t \geq 0$, where $v(x) = \dot{x}_d - f(x) - Ke$.

Let $\|\hat{\Lambda}(t)^{-1}\| \leq \kappa_{\hat{\Lambda}}$, where $\kappa_{\hat{\Lambda}} > 0$. Hence, the upper bounds on u_{nom} and $u_{\text{nom}}^{\text{FDI}}$ can be expressed by $\|u_{\text{nom}}\| \leq \kappa_g^\dagger \bar{v}$, and $\|u_{\text{nom}}^{\text{FDI}}\| = \|[g\hat{\Lambda}]^\dagger v\| \leq \kappa_{\hat{\Lambda}} \kappa_g^\dagger \bar{v}$.

Lemma 1: Under Assumption 3, one has $\|r\| \leq \bar{\beta}(\bar{u}_{\text{NN}} + \kappa_g^\dagger \bar{v}) + \|\hat{d}\| = \bar{r}_1$, and $\|r_{\text{FDI}}\| \leq \bar{\beta} \bar{u}_{\text{NN}} + \bar{\Delta}_\Lambda \kappa_{\hat{\Lambda}} \kappa_g^\dagger \bar{v} + \|\hat{d}\| = \bar{r}_2$, for all $t \geq 0$, where $\|\Lambda(t) - I\| \leq \bar{\beta}$ and $\bar{\beta} > 0$. Consequently, we obtain $\bar{r}_1 - \bar{r}_2 = \kappa_g^\dagger \bar{v}(\bar{\beta} - \kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda)$.

Proof: The proof readily follows from Assumption 3 and definitions of r and r_{FDI} . ■

From Lemma 1, it can be inferred that the difference between $\bar{r}_1$ and $\bar{r}_2$ depends on the accuracy of the FDI module in the fault estimation, i.e., $\|\Lambda - \hat{\Lambda}\|$, and consequently, its upper bound, i.e., $\bar{\Delta}_\Lambda$. Consider the adaptive law (4) and the backpropagated errors. Similar to Assumption 1, we consider an ideal representation of the residual r_{FDI} in the following assumption.

Assumption 4: Let $\ell^* \neq L$. There exists a bounded and constant weight W^* such that $r_{\text{FDI}} = W_L z_{L-1}^* + \epsilon_{\text{FDI}}(t)$, where z_{L-1}^* is generated by the DNN with W_{ℓ^*} replaced by W^* , $\|\epsilon_{\text{FDI}}(t)\| \leq \bar{\epsilon}_{\text{FDI}}$, and $\bar{\epsilon}_{\text{FDI}} > 0$. Moreover, if $\ell^* = L$, we have $r_{\text{FDI}} = W_L^* z_{L-1} + \epsilon_{\text{FDI}}(t)$.

Assumption 5: Let $\|r\| \leq \bar{r}_1$ and $\|r_{\text{FDI}}\| \leq \bar{r}_2$. Consider $\bar{\epsilon}$ and $\bar{\epsilon}_{\text{FDI}}$ from Assumptions 1 and 4, which are the upper bounds on the approximation errors achievable by u_{NN} when only the selected layer is adapted. There exists a constant $L_e > 0$ such that $\bar{r}_1 - \bar{r}_2 \leq L_e(\bar{\epsilon} - \bar{\epsilon}_{\text{FDI}})$.

Lemma 2: Under Assumptions 3 and 5, the approximation error bound of u_{NN} satisfies $\bar{\epsilon}_{\text{FDI}} \leq \bar{\epsilon} - \frac{1}{L_e} \kappa_g^\dagger \bar{v}(\bar{\beta} - \kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda)$, and one has a smaller approximation error when the FDI module fault estimation satisfies $\kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda < \bar{\beta}$.

Proof: In light of Lemma 1, the residual bound decreases by at least $\bar{r}_1 - \bar{r}_2 = \kappa_g^\dagger \bar{v}(\bar{\beta} - \kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda)$. Consequently, Assumption 5 yields $\bar{\epsilon}_{\text{FDI}} \leq \bar{\epsilon} - \frac{1}{L_e}(\bar{r}_1 - \bar{r}_2)$. ■

Using the same Lyapunov function in Section IV-A and derivations leading to (5), we obtain $\dot{V} = -e^\top Q e - \gamma \text{tr}(\tilde{W}^\top \Gamma^{-1} W_{\ell^*}) + e^\top P g \epsilon_{\text{FDI}} + \mathcal{O}$.

Theorem 2: Let Assumptions 2-5 hold. The closed-loop system with $u = u_{\text{nom}}^{\text{FDI}} + u_{\text{NN}}$ remains UUB with exponential convergence. Moreover, if the FDI estimates satisfy $\kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda < \bar{\beta}$, the upper bound on tracking error e and weight error $\tilde{W}$ will be smaller than the case of $u = u_{\text{nom}} + u_{\text{NN}}$ in Theorem 1.

Proof: The derivation to show $u = u_{\text{nom}}^{\text{FDI}} + u_{\text{NN}}$ results in having UUB follows along similar lines to that of Theorem 1. Moreover, it can be easily seen that the term $\sigma_{\text{FDI}} = \frac{\|P\|^2 \|g\|^2}{2\lambda_{\min}(Q)} \bar{\epsilon}_{\text{FDI}}^2 + \frac{\gamma}{2} \|\Gamma^{-\frac{1}{2}} W^*\|_F^2$ appears in the upper bounds for $\|e(t)\|$ and $\|\Gamma^{-1/2} \tilde{W}(t)\|_F$. Thus, considering Lemma 2, $\kappa_{\hat{\Lambda}} \bar{\Delta}_\Lambda < \bar{\beta}$ results in $\bar{\epsilon}_{\text{FDI}} < \bar{\epsilon}$, which implies smaller upper bounds for errors. ■

Theorem 2 explicitly quantifies how the FDI estimation accuracy, characterized by $\|\Lambda - \hat{\Lambda}\|$ and its upper bound $\bar{\Delta}_\Lambda$, affects the resulting tracking-error bound. Therefore, even in the presence of bounded but imperfect fault estimates, the closed-loop stability is maintained.

V. NUMERICAL CASE STUDY

To validate the proposed fault recovery control methodology, we consider a 3-axis attitude control system with loss of effectiveness faults for a rigid spacecraft that is actuated by four reaction wheels in a tetrahedral configuration. The inertia matrix is given by $I = \text{diag}(1, 1, 0.8)$ ($\text{kg} \cdot \text{m}^2$), each reaction wheel has an inertia of $J_\omega = 0.01$ ($\text{kg} \cdot \text{m}^2$), and the maximum deliverable torque per wheel is limited to 0.14 ($\text{N} \cdot \text{m}$). We have used [23] to define the spacecraft and reaction wheels parameters. The gains of the nominal controller u_{nom} are tuned as $K_p = \text{diag}(22.5, 18, 15)$ and $K_d = \text{diag}(12, 9, 7.5)$. Let $\theta = [\theta_1, \theta_2, \theta_3]^\top \in \mathbb{R}^3$ (rad) where θ_1 denotes the roll, θ_2 is the pitch, and θ_3 is the yaw angle of the spacecraft. The desired roll, pitch, and yaw angles are $\theta_d = [\theta_1^d, \theta_2^d, \theta_3^d]^\top$. We define the tracking errors as $\tilde{\theta}_1 = \theta_1 - \theta_1^d$, $\tilde{\theta}_2 = \theta_2 - \theta_2^d$, $\tilde{\theta}_3 = \theta_3 - \theta_3^d$, and $\tilde{\theta} = \theta - \theta_d$. The trained u_{NN} has 6 layers with the triple $(\tilde{\theta}, \dot{\tilde{\theta}}, u_{\text{nom}})$ as its input and layers of size $n_1 = 9$, $n_2 = 15$, $n_3 = 15$, $n_4 = 15$, $n_5 = 15$, and $n_6 = 3$ for the body-frame torque. Moreover, by utilizing $\widehat{\text{ACE}}_\ell$, we identified the 4-th layer as the best candidate to update in the recovery control. We consider 50% and 75% loss of effectiveness faults starting from $t = 10$ to 25 (s) and $t = 20$ to 50 (s) in reaction wheels number 3 and 2, respectively. Figure 2 illustrates $\|\tilde{\theta}\|$ while the 1-st, the 4-th, the 5-th, the 6-th (i.e., the last layer), the full-network (i.e., all the layers), and no layer of the network are being updated. As is shown in Fig. 2, updating the 4-th layer of u_{NN} results in having the smallest tracking error. Moreover, results in Table II show that updating the proposed ACE-selected layer (i.e., the 4-th layer) achieves the lowest root-mean-square error (RMSE) and maximum error (MaxError) values, with $\text{RMSE} = \sqrt{\frac{1}{T} \int_0^T \|\tilde{\theta}(t)\|^2 dt}$ and $\text{MaxError} = \max_{t \in [0, T]} \|\tilde{\theta}(t)\|$, for $T > 0$. Since the

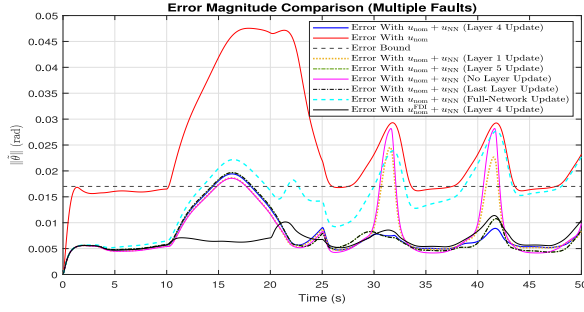


Fig. 2. Error $\|\tilde{\theta}\|$ (rad) while updating different u_{NN} layers.

TABLE II
COMPARISON OF TRACKING ERROR PERFORMANCE

Adaptation Strategy	RMSE (rad)	Max Error (rad)
Layer 4 (ACE-Selected)	0.008819	0.01943
Layer 5	0.008899	0.01967
Layer 6 (i.e., the last layer)	0.008899	0.01967
Layer 1	0.009898	0.02456
No Layer Update	0.010462	0.02819
Full-Network Update	0.015695	0.02769
Nominal Controller u_{nom}	0.027062	0.04755

computed ACE values $\widehat{ACE}_4 = -0.004387$ and $\widehat{ACE}_5 = \widehat{ACE}_6 = -0.004256$ are close, it is expected to get similar RMSEs for layers 4, 5, and 6 as shown in Table II. Finally, we incorporated an FDI module that provides fault estimates subject to estimation errors. As for the FDI module, we employ an Interacting Multiple-Model (IMM) estimator [24], where each filter corresponds to a certain actuator fault severity hypothesis. Under the same fault scenario a 40% loss of effectiveness was estimated for reaction wheel 3 (actual 50%), and a 45% loss was estimated for reaction wheel 2 (actual 75%). As illustrated in Fig. 2, the more accurate estimate for reaction wheel 3 leads to an improvement in tracking performance, whereas the less accurate estimate for reaction wheel 2 results in a degradation of performance. As for the computational complexity of the online phase, a forward pass through the DNN costs $O(\sum_{\ell=1}^L n_{\ell} n_{\ell-1})$. Backpropagating to a single adapted layer ℓ^* has cost $O(\sum_{\ell=\ell^*+1}^L n_{\ell} n_{\ell-1})$, and the update itself adds $O(n_{\ell^*} n_{\ell^*-1})$. In our network (9, 15, 15, 15, 15, 3), the total number of weights is 1080. However, in addition to backpropagation through layers 6 to 4, the ACE-selected layer 4 only has 225 parameters.

VI. CONCLUSION

This letter presented a fault recovery control framework for nonlinear control-affine systems with unknown actuator faults and disturbances. We introduced a two-stage methodology that evaluates the average causal effect (ACE) of each DNN layer to identify the most critical one for fault compensation and adapted only that layer online. The proposed causality-guided layer selection goes beyond the widely used last-layer adaptation strategy by revealing which layer is the most impactful for fault recovery control.

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